

Thermo Guardian:

Course: B.Tech CSE (First Year)

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Group Members:

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SUBMITTED To:

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1. Introduction

• Background & Context:

Temperature measurement plays a vital role in our daily lives and in various scientific, industrial, and environmental applications. Monitoring temperature is essential for maintaining safety, efficiency, and comfort in systems ranging from household appliances to large-scale manufacturing plants. With the rapid development of embedded systems and automation technologies, temperature detection and control have become more precise, efficient, and accessible.

The idea for this project arose from the need to design a simple, low-cost, and accurate temperature monitoring system using modern microcontroller technology. Traditional thermometers, although reliable, require manual observation and cannot provide continuous or automated data recording. To overcome this limitation, electronic temperature sensors integrated with microcontrollers like Arduino Uno offer a smart and automated solution.

In this project, we aim to design and implement a temperature sensor system capable of detecting and displaying real-time temperature readings. The Arduino Uno serves as the main controller, while sensors such as the LM35 or DHT11 are used to measure ambient temperature. The system can be expanded to trigger alerts or control devices based on specific temperature thresholds, making it applicable in smart homes, laboratories, and industrial setups.

This project also provides valuable hands-on experience in sensor interfacing, data processing, and embedded system programming, which are essential skills in modern electronics and automation engineering.

• Purpose & Objectives:

The main purpose of this project is to design and develop a temperature detection system that can accurately measure and display the surrounding temperature in real time. The system aims to provide an efficient, low-cost, and user-friendly solution for monitoring temperature using a temperature sensor and an Arduino Uno microcontroller. By implementing this project, we seek to understand the practical application of sensors and microcontrollers in automation systems and to demonstrate how temperature data can be collected, processed, and displayed for real-world use.

The specific objectives of the project are as follows:

1. To study and understand the working principles of temperature sensors such as LM35 or DHT11.
2. To interface the temperature sensor with the Arduino Uno microcontroller.
3. To design and implement a circuit that measures real-time temperature and displays it on an LCD screen or serial monitor.
4. To analyze the accuracy and reliability of the temperature readings obtained from the sensor.
5. To develop skills in programming, circuit designing, and sensor calibration.
6. To explore possible applications of temperature monitoring systems in daily life, such as smart homes, weather monitoring, and industrial automation.

- **Significance:**

The Temperature Sensor Project is significant because it addresses the growing need for accurate, affordable, and automated temperature monitoring systems in both everyday life and industrial applications. Temperature plays a vital role in controlling processes, ensuring safety, and maintaining comfort in various environments. Therefore, developing a system that can continuously measure and display temperature is both practical and educational.

This project demonstrates the real-world application of sensor technology and microcontroller-based automation. By using an Arduino Uno and a temperature sensor such as the LM35 or DHT11, the project provides an easy and effective method to collect and process environmental data. The system can be adapted for use in smart homes, laboratories, hospitals, weather stations, and industrial temperature monitoring, making it highly versatile.

From an academic perspective, the project enhances students' understanding of electronics, programming, and circuit design. It promotes hands-on learning, teamwork, and technical problem-solving skills, which are essential for engineering and technology students.

Overall, this project not only contributes to the development of a functional and low-cost temperature detection system but also strengthens the practical knowledge and creativity of the team members.

2. Research & Planning

- **Initial Research:**

The initial research for our project focused on understanding the fundamental principles, components, and applications of temperature sensing systems. Temperature is one of the most important physical parameters in daily life and industry; therefore, accurate detection and monitoring are essential in various fields such as environmental control, healthcare, and automation.

We began by studying the concept and working principle of temperature sensors. A temperature sensor is a device that measures the degree of hotness or coldness of an environment or object and converts this information into an electrical signal. Several types of sensors were explored, including thermistors, thermocouples, LM35, and DHT11. Among these, the LM35 sensor and DHT11 were found to be the most suitable for small-scale projects because of their low cost, simple interfacing, and reliable accuracy.

Next, we researched how to interface these sensors with a microcontroller, specifically the Arduino Uno, which acts as the processing unit. The Arduino reads the analog or digital signals from the sensor and converts them into temperature readings in Celsius or Fahrenheit. We studied the use of the Arduino IDE for coding and data visualization on a serial monitor or LCD display.

Our literature review also covered existing applications of temperature sensors, such as in air conditioners, weather monitoring systems, industrial machines, and smart home devices. This helped us understand how temperature monitoring contributes to safety, comfort, and energy efficiency.

Finally, we analyzed the components and tools required for implementation. These include the Arduino Uno board, LM35/DHT11 sensor, jumper wires, a breadboard, and a display unit. We also studied power requirements and data acquisition methods to ensure accurate and stable readings.

Through this research, we developed a strong understanding of both the theoretical and practical aspects of temperature sensing technology. This knowledge laid the foundation for designing and implementing an efficient and reliable temperature detection system..

• Planning & Strategy:

Effective planning and strategy were essential to ensure the successful completion of our Temperature Sensor Project. Since the project was completed by a group of four members — Arjit, Avishkar, Ayush Badoni, and Ayush Dhasmana — we divided our work equally to make sure every stage of development was properly managed and executed.

The project was divided into several key phases to ensure smooth progress and proper time management:

Phase	Task Description	Duration
Phase 1: Research & Conceptualization	Studying different temperature sensors, understanding Arduino working, and defining objectives.	Week 1
Phase 2: Component Selection	Choosing suitable components such as Arduino Uno, LM35/DHT11 sensor, jumper wires, and LCD display	Week 2
Phase 3: Circuit Design & Assembly	Building the circuit on a breadboard and connecting the components properly.	Week 3
Phase 4: Coding & Testing	Writing Arduino code, uploading it to the board, and testing for accurate temperature readings.	Week 4
Phase 5: Documentation & Final Review	Preparing the project report, presentation, and demonstration setup.	Week 5

• Roles & Responsibilities:

Member	Role	Responsibility
Arjit Chauhan	Team Leader / Planner	Coordinate the team, assign tasks, manage timeline, and ensure progress.

Ayush Badoni	Hardware Lead	Set up sensors (like LM35 or DHT11), connect Arduino, manage wiring and circuit.
Avishkar Rawat	Software & Coding lead	Write and debug Arduino code for reading and displaying temperature data.
Ayush Dhasmana	Documentation & Presentation Lead	Prepare report, collect data, make charts, and design PowerPoint presentation.

3. Project Development & Process

• Activities & Steps Taken:

While making this project, we had to go through many different stages. First step was to create the basic circuit diagram for our model. Then came the initialization part where we gathered the materials required and assembled them together (the Arduino board, display, heat sensor) using jumper wires. Then came the most exciting part of all, the code which was written in C on the Arduino IDE software. With this, we also had to constantly debug and run multiple test cases. After which finally our project was completed. But wait that's not all we also had to make it look presentable, so we used cardboard, sketch pens, coloured sheets to complete it.

• Challenges & Solutions:

While completing this project we had several setbacks which were gradually resolved. The very first issue we faced was that the program we created had loopholes and errors which were so small to detect. After constant debugging sessions, we finally managed to make it work and modified it several times. The second major setback was due to our display not working. This led us to check every circuit again and we finally found the loophole in the code which was making the output to be displayed on wrong pin.

• Tools & Resources Used:

The successful implementation of the temperature sensor project required a combination of hardware components, software tools, and other supporting materials. Each tool and resource played an important role in building, testing, and operating the system effectively.

1. Hardware Components

Arduino Uno: Acts as the main microcontroller for processing and controlling the system.

Temperature Sensor (LM35 or DHT11): Measures the ambient temperature and provides output to the Arduino.

Breadboard: Used for circuit assembly during testing and prototyping.

Jumper Wires: Connect different components within the circuit.

16×2 LCD Display / Serial Monitor: Displays the temperature readings in real time.

Resistors and Potentiometer: Used for controlling current and adjusting display contrast.

USB Cable and Power Supply: Provide power and allow data transfer to the Arduino.

2. Software Tools

Arduino IDE: Used for writing, compiling, and uploading the program to the Arduino Uno.

C/C++ Programming Language: For coding logic and sensor data processing.

Serial Monitor (in Arduino IDE): For displaying temperature readings during testing and debugging.

3. Additional Resources

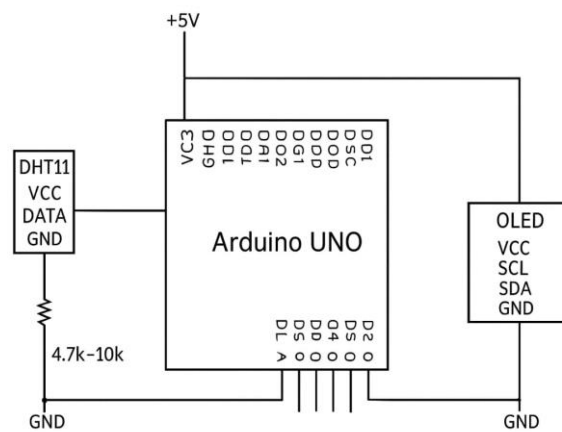
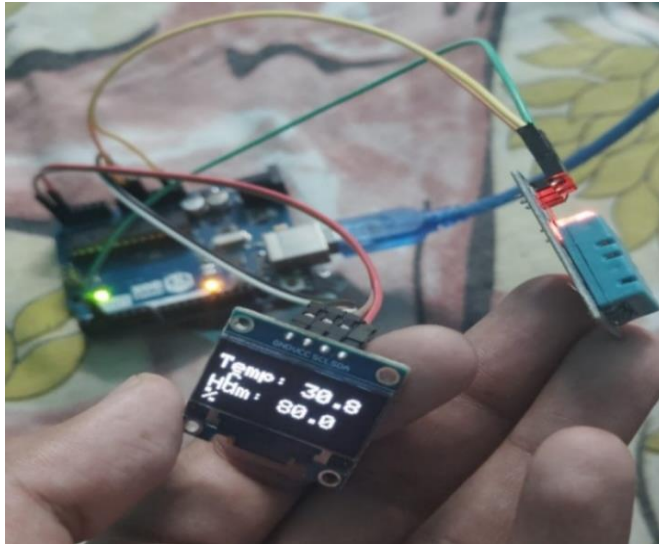
Research papers, online tutorials, and Arduino documentation for technical reference.

Team collaboration tools for report preparation and data collection.

4. Results & Findings

• Data & Evidence:

Result of our first working test case (when we tested it on heat):-



- **Analysis:**

We noted several readings and checked the working conditions for the project to work thus leading us into a deeper understanding of our project and how sensors work.

- **Discussion:**

At the end, we were happy that our project was up to our expectations, and we believe that it turned out to be less costly from what we expected.

5. Conclusion & Reflection

- **Summary of Outcomes:**

The project showed that temperature can be measured accurately using thermometers and sensors. Digital sensors were found to give faster and more precise readings. It also highlighted the importance of calibration and environmental factors in measurement. Overall, the project improved understanding of temperature measurement and its real-life uses.

- **Learning & Skills Gained:**

We learned how temperature is measured using devices and sensors. The project improved our skills in data recording, observation, and analysis. It also helped to understand real-world applications of temperature measurement and develop basic problem-solving abilities.

- **Future Recommendations:**

In the future, more advanced digital or IoT-based sensors can be used for higher accuracy. Automatic data recording and monitoring systems should be added. Further research can help improve calibration, efficiency, and real-time temperature tracking.

6. References

https://en.wikipedia.org/wiki/Temperature_measurement